

FINAL PROJECT – Individual Report [Week 2]

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Group Number: Group 3

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Calendar Week: CW41

Date: [Oct 13–19, 2025]

Week 1 Progress:

- Created CAD files for field objects (Ball and Cube), Electronic parts (Servo and DC motor), and other additional parts.
- Created CAD file for parts that required a 3D printer to be printed and start printing.
- Completed the design on V1 Auto Robot.
- Start designing the Manual robot by separating into different parts according to the functions. (Arms, Drive frame, Launcher)

Challenges & Solutions:

- **Challenge 1:** While designing the Auto Robot, the rulebook specification on size limit causes the design problem when assembling the cube collecting mechanism to the drive frame.
- **Solution:** I have made an objective cube in CAD file to compare to the size of the robot and find the most optimal position for the gate mechanism to be placed on.
- **Challenge 2:** With shared metal parts between the Auto and Manual robots, there is a limit in the number and type of component that can be used which also affects the designing process of the Manual robot.
- **Solution:** Instead of designing on CAD, we try to assemble with real components left from the Auto robot which visualize how many parts left we got for the Manual robot.

Next Week Plan:

- Finished the Auto robot design and assembled it with real components.
- Test the sensors and motors of the Auto robot.
- Complete the Manual robot design and print the required part for the assembly process.